(ii) To minimize the negative values like, expenses, efforts etc.,

(iii) Used in the study of inventory control, economic order quantity etc.

6.2.1 Increasing and decreasing functions

Before learning the concept of maxima and minima, we will study the nature of the curve of a given function using derivative.

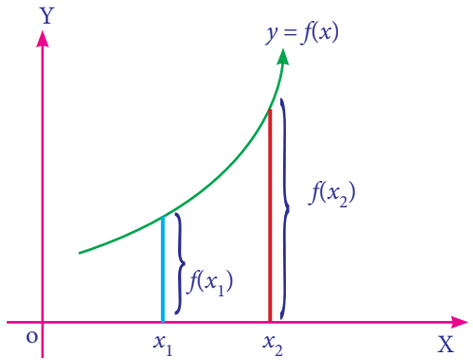
(i) Increasing function

A function $f(x)$ is said to be increasing function in the interval $[a,b]$ if

$$x_1 < x_2 \Rightarrow f\left(x_1\right) \leq f\left(x_2\right) \text{ for all } x_1, x_2 \in \left[a,b\right]$$

A function $f(x)$ is said to be strictly increasing in $[a,b]$ if

$$x_1 < x_2 \Rightarrow f\left(x_1\right) < f\left(x_2\right) \text{ for all } x_1, x_2 \in \left[a,b\right]$$



$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$

Strictly increasing function

Fig: 6.7

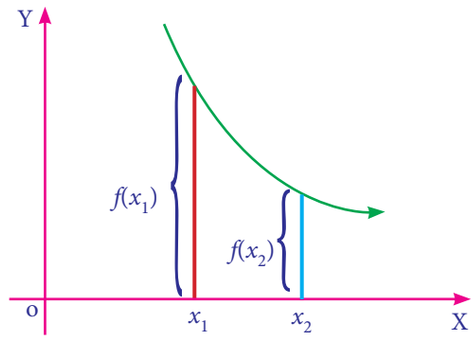
(ii) Decreasing function

A function $f(x)$ is said to be decreasing function in $[a, b]$ if

$$x_1 < x_2 \Rightarrow f\left(x_1\right) \geq f\left(x_2\right) \text{ for all } x_1, x_2 \in \left[a,b\right]$$

A function $f(x)$ is said to be strictly decreasing function in $[a,b]$ if

$$x_1 < x_2 \Rightarrow f\left(x_1\right) > f\left(x_2\right) \text{ for all } x_1, x_2 \in \left[a,b\right]$$



$x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$

Strictly decreasing function

Fig: 6.8

NOTE

A function is said to be monotonic function if it is either an increasing function or a decreasing function.

Derivative test for increasing and decreasing function

Theorem: 6.1 (Without Proof)

Let $f(x)$ be a continuous function on $[a,b]$ and differentiable on the open interval (a,b), then

(i) $f(x)$ is increasing in $[a, b]$ if $f'\left(x\right) \geq 0$

(ii) $f(x)$ is decreasing in $[a, b]$ if $f'\left(x\right) \leq 0$

Remarks:

(i) $f(x)$ is strictly increasing in (a,b) if $f'\left(x\right) > 0$ for every $x \in \left(a,b\right)$

(ii) $f(x)$ is strictly decreasing in (a,b) if $f'\left(x\right) < 0$ for every $x \in \left(a,b\right)$

(iii) $f(x)$ is said to be a constant function if $f'\left(x\right) = 0$

6.2.2 Stationary Value of a function

Let $f(x)$ be a continuous function on $[a, b]$ and differentiable in (a, b). $f(x)$ is said to be stationary at $x = a$ if $f'(a)=0$.

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